

Database Design and Sample Data Report

Project Title

Smart Timetable Generator Portal

1. Overview

The Smart Timetable Generator Portal uses **MongoDB Atlas** as its primary NoSQL database. The database is designed to support timetable generation, user authentication, department management, faculty allocation, classroom scheduling, and institutional reporting.

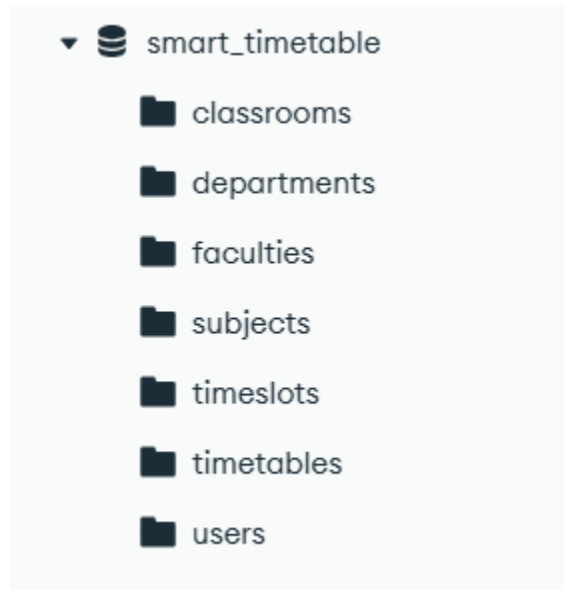
The system follows a **rule-based scheduling approach** and does not use any Artificial Intelligence or Machine Learning techniques.

MongoDB collections are connected using **Mongoose ObjectId references**, enabling efficient relationships between related entities while maintaining the flexibility of a document database.

2. Database Collections

Collection	Purpose
Users	Stores login credentials and role information
Departments	Stores department details
Faculty	Stores faculty information and department mapping
Subjects	Stores academic subject information
Classrooms	Stores classroom details
TimeSlots	Stores available teaching time slots

Timetables Stores generated timetable entries



3. Collection Schemas

3.1 Users Collection

Purpose

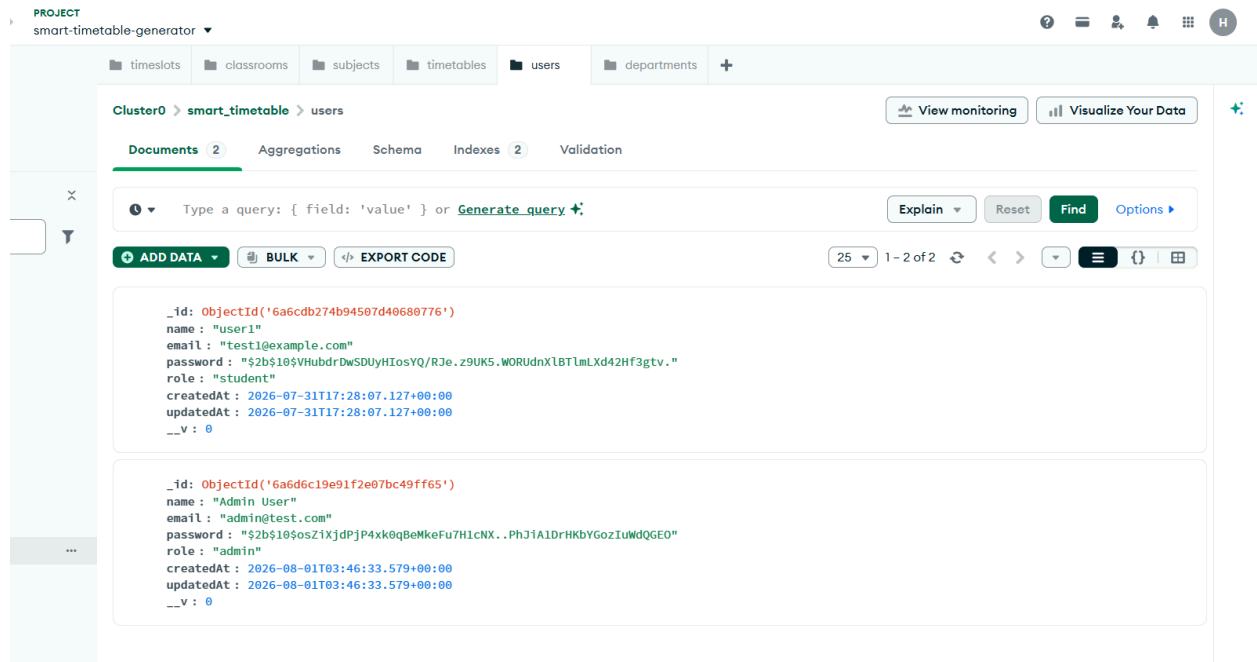
Stores authenticated users of the system.

Schema

Field	Data Type	Description
name	String	User's full name
email	String	Unique email address
password	String	Encrypted password using bcrypt
role	Enum	admin / faculty / student

Sample Document

```
{
  "name": "Admin User",
  "email": "admin@college.edu",
  "password": "Encrypted Password",
  "role": "admin"
}
```



3.2 Departments Collection

Purpose

Stores academic departments.

Schema

Field	Type
name	String
code	String
hod	String
status	Active / Inactive

Sample Documents

```
{
  "name": "Computer Science",
  "code": "CSE",
  "hod": "Dr. Ramesh",
  "status": "Active"
}
```

```
{
  "name": "Information Technology",
  "code": "IT",
  "status": "Active"
}
```

```
{
  "name": "Electronics and Communication",
  "code": "ECE",
  "status": "Active"
}
```

Cluster0 > smart_timetable > departments

Documents 1 Aggregations Schema Indexes 3 Validation

Type a query: { field: 'value' } or [Generate query](#)

Explain Reset Find Options

ADD DATA BULK EXPORT CODE

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```
{
  "_id": ObjectId('6a6d6c7fe91f2e07bc49ff66'),
  "name": "Computer Science",
  "code": "CSE",
  "createdAt": 2026-08-01T03:48:15.858+00:00,
  "updatedAt": 2026-08-01T03:48:15.858+00:00,
  "__v": 0
}
```

3.3 Faculty Collection

Purpose

Stores faculty members and their department associations.

Schema

Field	Type
name	String

email	String
phone	String
department	ObjectId (Department)
designation	String
subjectsHandled	Array
d	
isAvailable	Boolean

Sample Documents

```
{  
  "name": "Dr. Ananya Rao",  
  "email": "ananya@college.edu",  
  "phone": "+91 9876543210",  
  "department": "<IT Department ID>",  
  "designation": "Professor",  
  "subjectsHandled": [],  
  "isAvailable": true  
}
```

```
{  
  "name": "Dr. Arjun Kumar",  
  "email": "arjun@college.edu",  
  "phone": "+91 9870012345",  
  "department": "<ECE Department ID>",  
  "designation": "Associate Professor",  
  "subjectsHandled": [],  
  "isAvailable": true  
}
```

Cluster0 > smart_timetable > faculties

View monitoring

Visualize Your Data

Documents 1

Aggregations

Schema

Indexes 2

Validation

Type a query: { field: 'value' } or [Generate query](#)

Explain

Reset

Find

Options

ADD DATA

BULK

EXPORT CODE

25

1 - 1 of 1

```
_id: ObjectId('6a6d6edcb2c99010a5102f02')
name: "Dr. Kavitha"
email: "kavitha@college.edu"
department: ObjectId('6a6d6c7fe91f2e07bc49ff66')
designation: "Assistant Professor"
subjectsHandled: Array (empty)
isAvailable: true
createdAt: 2026-08-01T03:58:20.580+00:00
updatedAt: 2026-08-01T03:58:20.580+00:00
__v: 0
```

3.4 Subjects Collection

Purpose

Stores all academic subjects.

Schema

Field	Type
name	String
code	String
department	ObjectId
semester	Number
credits	Number

Sample Documents

```
{
  "name": "Operating Systems",
  "code": "CS301",
  "department": "<CSE Department ID>",
  "semester": 3,
  "credits": 4
}
```

```

}

{
  "name": "Digital Electronics",
  "code": "ECE201",
  "department": "<ECE Department ID>",
  "semester": 3,
  "credits": 3
}

```

The screenshot shows the MongoDB Compass interface. At the top, the breadcrumb navigation is 'Cluster0 > smart_timetable > subjects'. There are two buttons: 'View monitoring' and 'Visualize Your Data'. Below this, the 'Documents' tab is selected, showing '1' document. Other tabs include 'Aggregations', 'Schema', 'Indexes' (showing '2'), and 'Validation'. A search bar contains the text 'Type a query: { field: 'value' } or [Generate query](#)'. To the right of the search bar are buttons for 'Explain', 'Reset', 'Find', and 'Options'. Below the search bar, there are buttons for 'ADD DATA', 'BULK', and 'EXPORT CODE'. On the right side of this bar, there is a dropdown set to '25', a '1 - 1 of 1' indicator, and navigation icons. The main area displays a single document in JSON format:

```

{
  "_id": ObjectId('6a6d72d54fd0f261c0e0c05b'),
  "name": "Data Structures",
  "code": "CS201",
  "department": ObjectId('6a6d6c7fe91f2e07bc49ff66'),
  "semester": 3,
  "credits": 4,
  "createdAt": 2026-08-01T04:15:17.780+00:00,
  "updatedAt": 2026-08-01T04:15:17.780+00:00,
  "__v": 0
}

```

3.5 Classrooms Collection

Purpose

Stores classroom and laboratory information.

Schema

Field	Type
roomNumber	String
capacity	Number
type	lecture / lab / seminar

Sample Documents

```

{
  "roomNumber": "Room-101",
  "capacity": 60,
  "type": "lecture"
}

{
  "roomNumber": "Lab-201",
  "capacity": 30,
  "type": "lab"
}

{
  "roomNumber": "Seminar-Hall-1",
  "capacity": 100,
  "type": "seminar"
}

```

The screenshot shows the MongoDB Atlas web interface. At the top, the breadcrumb navigation indicates the path: Cluster0 > smart_timetable > classrooms. There are two buttons: 'View monitoring' and 'Visualize Your Data'. Below the breadcrumb, there are tabs for 'Documents' (selected), 'Aggregations', 'Schema', 'Indexes' (2), and 'Validation'. A search bar contains the text 'Type a query: { field: 'value' } or Generate query'. Below the search bar, there are buttons for 'ADD DATA', 'BULK', and 'EXPORT CODE'. The main area displays a single document with the following fields: `_id` (ObjectId), `roomNumber` (string), `capacity` (number), `type` (string), `createdAt` (timestamp), `updatedAt` (timestamp), and `__v` (version number).

3.6 TimeSlots Collection

Purpose

Defines available teaching periods.

Schema

Field	Type
day	Enum
startTime	String

endTime String

Sample Documents

```
{
  "day": "Monday",
  "startTime": "09:00",
  "endTime": "10:00"
}
```

```
{
  "day": "Tuesday",
  "startTime": "10:00",
  "endTime": "11:00"
}
```

```
{
  "day": "Wednesday",
  "startTime": "11:00",
  "endTime": "12:00"
}
```

Cluster0 > smart_timetable > timeslots

View monitoring Visualize Your Data

Documents 1 Aggregations Schema Indexes 1 Validation

Type a query: { field: 'value' } or [Generate query](#)

Explain Reset Find Options

ADD DATA BULK EXPORT CODE

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```
{
  "_id": ObjectId('6a6d7541b4bb9bcc42d907b9'),
  "day": "Monday",
  "startTime": "09:00",
  "endTime": "10:00",
  "createdAt": 2026-08-01T04:25:38.000+00:00,
  "updatedAt": 2026-08-01T04:25:38.000+00:00,
  "__v": 0
}
```

3.7 Timetables Collection

Purpose

Stores timetable entries generated by the scheduling module.

Schema

Field	Type
department	ObjectId
semester	Number
section	String
subject	ObjectId
faculty	ObjectId
classroom	ObjectId
timeSlot	ObjectId
academicYear	String
status	active / cancelled / rescheduled

Sample Document

```
{
  "department": "<Department ID>",
  "semester": 3,
  "section": "A",
  "subject": "<Subject ID>",
  "faculty": "<Faculty ID>",
  "classroom": "<Classroom ID>",
  "timeSlot": "<TimeSlot ID>",
  "academicYear": "2026-2027",
  "status": "active"
}
```

Cluster0 > smart_timetable > timetables

View monitoring
Visualize Your Data

Documents 1
Aggregations
Schema
Indexes 1
Validation

Type a query: { field: 'value' } or [Generate query](#)

Explain
Reset
Find
Options

ADD DATA
BULK
EXPORT CODE

25
1 - 1 of 1

```

_id: ObjectId('6a6d8a27124b6bd34988e863')
department: ObjectId('6a6d6c7fe91f2e07bc49ff66')
semester: 3
section: "A"
subject: ObjectId('6a6d72d54fd0f261c0e0c05b')
faculty: ObjectId('6a6d6edcb2c99010a5102f02')
classroom: ObjectId('6a6d73464fd0f261c0e0c05c')
timeSlot: ObjectId('6a6d7541b4bb9bcc42d907b9')
academicYear: "2026-2027"
status: "active"
createdAt: 2026-08-01T05:54:47.668+00:00
updatedAt: 2026-08-01T05:54:47.668+00:00
__v: 0

```

4. Collection Relationships

The database maintains relationships through MongoDB ObjectId references.

- A **Department** can have multiple Faculty members.
- A **Department** offers multiple Subjects.
- A **Faculty** member can teach multiple Subjects.
- A **TimeSlot** can be assigned to multiple timetable entries across different days and sections.
- A **Classroom** is allocated to timetable entries based on availability and capacity.
- A **Timetable** entry references a Department, Subject, Faculty, Classroom, and TimeSlot.
- A **User** represents authenticated system users with role-based access (Admin, Faculty, or Student).

5. Sample Dataset Summary

Collection	Sample Records
Users	3
Department s	3
Faculty	2

Subjects	2
Classrooms	3
TimeSlots	3
Timetables	1

6. Database Features

- MongoDB Atlas cloud-hosted database
 - Mongoose ODM for schema modeling
 - ObjectId-based relationships between collections
 - Secure password storage using bcrypt hashing
 - Role-based user management
 - Scalable document-oriented design
 - Optimized for timetable generation and institutional scheduling
 - Extensible structure for future modules such as notifications, reports, guest lectures, and substitute faculty management
-

7. Conclusion

The proposed MongoDB database schema provides a scalable, modular, and maintainable foundation for the Smart Timetable Generator Portal. The use of referenced collections minimizes data redundancy while enabling efficient retrieval of related information for timetable generation, faculty allocation, classroom management, and administrative reporting. The design is well-suited for the project's rule-based scheduling approach and supports future expansion without requiring significant structural changes.